

Functional Requirements

1. UVUSim will read the language BasicML and interpret it to execute the corresponding operations
2. UVUSim will allow the user to upload their instructions to the file if the file is in a .txt format and the instructions are in number format with a + or - in front of them.
3. UVUSim will halt when the instruction is 4300
4. UVUSim will save the instructions in memory starting from address 00
6. UVUSim will iterate through each instruction in memory.
7. UVUSim can manipulate the accumulator if the instructions require it to be compared or changed.
8. UVUSim will display an error message if the user input is not an integer 0- 9.
11. UVUSim will be able to branch to other locations of memory when given the instruction 40.
12. UVUSim will be able to branch to other locations of memory depending on the value of the accumulator if the instructions are 41 or 42.
13. UVUSim will be able to write and read inputs from the user/memory.
14. UVUSim will wipe its memory once a new set of instructions is executed.
15. UVSim's GUI should have a button that begins the execution of the loaded program.

Non-Functional Requirments

1. The program will be able to handle instructions that can be up to 100 instructions in length
2. The program will be able to run the instructions and execute them within .5 seconds.
3. The input and output of UVSim's CPU should be decoupled from the CPU's implementation. (i.e., the CPU should be able to be used with any source of input.)